

# Dooyoung Kim

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## CURRENT APPOINTMENT

From Jun 2026 **Lecturer in Computer Science and Information Technology**,  
School of Computing, Engineering and Mathematical Sciences,  
La Trobe University, Melbourne, Australia.  
Continuing academic appointment.

## RESEARCH INTERESTS

Areas AR/VR, human-computer interaction, interactive media, XRMemory, digital replica; interactive XR intelligence; spatial AI, XR for physical AI.  
Direction *Connecting intelligences beyond boundaries*: building XR systems that connect human perception, memory, action, and collaboration with AI agents, robots, and spatial environments.

## EDUCATION

Mar 2021 – Feb 2024 **Ph.D. in Culture Technology**, KAIST, Daejeon, Korea.  
Thesis: *Space-Adaptive Mutual Space Generation for Mixed Reality Remote Collaboration*.  
UVR Lab. · Advisor: Prof. Woontack Woo.  
Visiting Researcher (2020): Imagine Lab, École des Ponts ParisTech, France (Prof. Vincent Lepetit).  
Mar 2019 – Feb 2021 **M.S. in Culture Technology**, KAIST, Daejeon, Korea.  
Thesis: *Adjusting Relative Translation Gains According to Space Size in Redirected Walking for Mixed Reality Mutual Space Generation*.  
UVR Lab. · Advisor: Prof. Woontack Woo.  
Mar 2014 – Feb 2019 **B.S. in Mechanical Engineering**, KAIST, Daejeon, Korea.  
Exchange Student (2016): Tsinghua University, Beijing, China.  
Lab Interns: Hubo Lab — Humanoid Robotics (2017); UVR Lab — AR/VR (2018).

## PREVIOUS APPOINTMENTS

Mar 2024 – May 2026 **Senior Researcher**, KAIST, Daejeon, Korea.  
Co-PI on the \$3M Meta-Object national project; System Integration Manager on the \$6M OpenXR platform. Supervised 7 graduate students (PhD/MS), with publications at IEEE TVCG, ISMAR, VR.  
Jan 2025 – Dec 2025 **NYU–KAIST Postdoc Collaboration**, New York University, NY, USA.  
NYU Future Reality Lab (Prof. Ken Perlin); on-site and remote visits throughout 2025 on the KAIST–NYU Meta-Museum initiative.  
May 2022 – Mar 2023 **AR/VR Researcher**, Korea Institute of Science and Technology Information (KISTI), Daejeon, Korea.  
Joint research with KAIST on the OpenXR: Untact Realistic Platform Technology government project.  
Apr 2021 – Nov 2022 **Creator**, ZER01NE program, Hyundai Motors Group (HMG).  
Selected as AR/VR researcher. Projects: *Automatic SONATA* — scenery style-transfer system for autonomous vehicles (2021); *Holo-bot* — XR social interface robot with holographic avatar (2022).

## TECHNICAL EXPERTISE

Platforms Meta Quest 3/Pro, HoloLens 2, Apple Vision Pro, custom CAVE/wall-display setups, OpenXR.

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| Software | Unity (C#), Unreal, ARKit/ARCore, MRTK; Python (PyTorch, NumPy); OpenCV, COLMAP/SLAM; human-subject statistics.                                                                                                           |
| Methods  | XR prototyping, spatial AI / scene understanding, semantic scene graphs, 3D reconstruction, world-model priors, redirected walking, perceptual experiments, IRB-approved human-subject studies, mixed-methods evaluation. |

## AWARDS AND HONOURS

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| Oct 2025 | <b>Best Paper Award</b> (Top 1%), IEEE ISMAR 2025 — <i>Viewpoint-Tolerant Depth Perception for Shared Extended Space Experience on Wall-Sized Display</i> .                      |
| Oct 2025 | <b>Best Paper Award</b> (Top 1%), IEEE ISMAR 2025 — <i>Visuo-Tactile Feedback with Hand Outline Styles for Modulating Affective Roughness Perception</i> .                       |
| Jan 2025 | NYU–KAIST Postdoc Collaboration Grant, New York University.                                                                                                                      |
| Oct 2024 | <b>Best Paper Award</b> (Top 1%), IEEE ISMAR 2024 — <i>Spatial Affordance-Aware Interactable Subspace Allocation for Mixed Reality Telepresence</i> .                            |
| Mar 2023 | KAIST Start-up Scholarship, KAIST Alumni Academic Scholarship Foundation.                                                                                                        |
| Dec 2022 | <b>Best Presentation Award</b> (1st Prize), The Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR) 2022 — Springer <i>Virtual Reality</i> paper invited and presented. |
| Apr 2022 | NCSOFT Ph.D. Scholarship.                                                                                                                                                        |
| May 2021 | <b>Honorable Mention Award</b> (Top 5%), ACM CHI 2021.                                                                                                                           |

## GRANTS AND RESEARCH PROJECTS

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|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jan 2025 – Dec 2025 | <b>Meta-Museum</b> — NYU–KAIST Collaborative Research Grant.<br><i>Tele-experience system for a meta-museum between Daejeon (Korea) and New York (USA)</i> .<br><b>Role:</b> Project Leader. Collaboration with NYU Future Reality Lab (Prof. Ken Perlin).                                                                                                                                      |
| Apr 2024 – May 2026 | <b>Meta-Object</b> — Korean Government (IITP) national research project.<br><i>Real-time XR interface technology for environmental adaptation; object modelling, multimodal feedback, intelligent platform</i> .<br><b>Role:</b> Co-Principal Investigator (Co-PI). Led the successful grant proposal securing <b>\$3M</b> ; directing <b>40 researchers across 8 multidisciplinary teams</b> . |
| Dec 2021 – May 2026 | <b>OpenXR: TranSpace 3.0</b> — Korean Government (NST) national research project.<br><i>Development of untact realistic OpenXR remote collaboration platform technology</i> .<br><b>Role:</b> System Integration Manager. Implementation lead for the KAIST team. <b>\$6M over 6 years</b> .                                                                                                    |
| Sep 2019 – Dec 2020 | <b>TranSpace 2.0</b> — Korean Government (NRF) national research project.<br><i>User restoration technology for ultra-realistic remote virtual interaction</i> .<br><b>Role:</b> Research Assistant. AR/VR remote-collaboration system and VR locomotion research.                                                                                                                              |
| Apr 2020 – Dec 2020 | <b>National R&amp;D Real Challenge</b> — Korean Government (KIRD).<br><i>Reinforcement of visual cognitive abilities for AR/VR/MR remote collaboration</i> .<br><b>Role:</b> Research Assistant. Visual-perception research for VR remote-collaboration scenarios.                                                                                                                              |

## PUBLICATIONS

|         |                                                                                                                                                                                                       |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Summary | <b>31 refereed publications</b> in total — <b>12 journal</b> ([J.1–J.12]), <b>8 conference</b> ([C.1–C.8]), and <b>11 workshop / adjunct / demo</b> ([W.1–W.11]). All listed below are peer-reviewed. |
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*Bold name indicates author; ★ marks award-winning papers. Co-advised students' first-author papers are labelled (co-advised, student first author).*

### Refereed journal papers

- [J.12] S.-Y. Kim, **Dooyoung Kim**, T. Son, Y. B. Kim, W. Cho, W. Woo. “Mix3R: Zero-shot Interactive 3D Scene Reconstruction from a Single Image in Mixed Reality.” *IEEE TVCG*, 2026 (accepted). (co-advised, student first author)
- [J.11] S. Kim, **Dooyoung Kim**, S. Choi, W. Woo. “Spatial Affordance-Aware Affine Transformation Between Heterogeneous Spaces for Mixed Reality Remote Collaboration.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.10] S.-Y. Kim, **Dooyoung Kim**, W. Choi, H. Song, S. Kang, W. Woo. “SceneLinker: Compositional 3D Scene Generation via Semantic Scene Graph from RGB Sequences.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.9] S. Choi, **Dooyoung Kim**, T. Ha, S. Kim, W. Woo. “Task Breakpoint Generation using Origin-Centric Graph in Virtual Reality Recordings for Adaptive Playback.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.8] T. Ha, W. Choi, **Dooyoung Kim**, W. Woo. “Int3DNet: Scene-Motion Cross-Attention Network for 3D Intention Prediction in Mixed Reality.” *IEEE TVCG*, 2026. (co-advised, student first author)
- [J.7] **Dooyoung Kim**, J. Hong, H. Ko, W. Woo. “Viewpoint-Tolerant Depth Perception for Shared Extended Space Experience on Wall-Sized Display.” *IEEE TVCG*, 2025. ★ **Best Paper Award (Top 1%)**, **IEEE ISMAR 2025**
- [J.6] M. Baeck, Y. Shin, **Dooyoung Kim**, H. Lee, S. H. Yoon, W. Woo. “Visuo-Tactile Feedback with Hand Outline Styles for Modulating Affective Roughness Perception.” *IEEE TVCG*, 2025. (co-advised, student first author) ★ **Best Paper Award (Top 1%)**, **IEEE ISMAR 2025**
- [J.5] **Dooyoung Kim**, T. Ha, J. Hong, S. Kim, S. Choi, H. Ko, W. Woo. “Meta-Objects: Interactive and Multisensory Virtual Objects Learned from the Real World for Use in Augmented Reality.” *IEEE Computer Graphics and Applications*, 2025.
- [J.4] **Dooyoung Kim**, S. Kim, J.-e. Shin, B. Yoon, J. Kim, J. Lee, W. Woo. “The Effects of Spatial Configuration on Relative Translation Gain Thresholds in Redirected Walking.” *Springer Virtual Reality*, 2023. ★ **Best Presentation Award (1st Prize)**, **APMAR 2022**
- [J.3] J. Kim, H. Jang, **Dooyoung Kim**, J. Lee. “Exploration of the VR Teleportation Methods Using Hand Tracking, Eye Tracking, and EEG.” *IJHCI*, 2023.
- [J.2] B. Yoon, J.-e. Shin, H.-i. Kim, S. Oh, **Dooyoung Kim**, W. Woo. “Effects of Avatar Transparency on Social Presence in Task-Centric Mixed Reality Remote Collaboration.” *IEEE TVCG*, 2023.
- [J.1] J.-e. Shin, B. Yoon, **Dooyoung Kim**, W. Woo. “The Effects of Spatial Complexity on Narrative Experience in Space-Adaptive AR Storytelling.” *IEEE TVCG*, 2022.

### Refereed conference papers

- [C.8] S.-Y. Kim, A. Elskhawy, T. Ha, **Dooyoung Kim**, E. Shin, B. Busam, W. Woo. “DepWorldSG: Depth-Aware 3D Semantic Scene Graph Generation via World-Model Priors.” *ECCV*, 2026 (accepted). (co-advised, student first author)
- [C.7] S. Choi, S. A. Bae, **Dooyoung Kim**, W. Woo. “Object-Driven Event Retrieval with Assembled Object Interface for Mixed-Reality Assembly Playback.” *IEEE ISMAR*, 2026 (accepted). (co-advised, student first author)
- [C.6] **Dooyoung Kim**, S. Kim, S. Choi, W. Woo. “Spatial Affordance-Aware Interactable Subspace Allocation for Mixed Reality Telepresence.” *IEEE ISMAR*, 2024. ★ **Best Paper Award (Top 1%)**
- [C.5] **Dooyoung Kim**, W. Woo. “Edge-Centric Space Rescaling with Redirected Walking for Dissimilar Physical-Virtual Space Registration.” *IEEE ISMAR*, 2023.
- [C.4] **Dooyoung Kim**, J. Kim, J.-e. Shin, B. Yoon, J. Lee, W. Woo. “Effects of Virtual Room Size and Objects on Relative Translation Gain Thresholds in Redirected Walking.” *IEEE VR*, 2022.

- [C.3] J.-e. Shin, B. Yoon, **Dooyoung Kim**, H.-i. Kim, W. Woo. “The Effects of Device and Spatial Layout on Social Presence during a Dynamic Remote Collaboration Task in Mixed Reality.” *IEEE ISMAR*, 2022.
- [C.2] **Dooyoung Kim**, J.-e. Shin, J. Lee, W. Woo. “Adjusting Relative Translation Gains According to Space Size in Redirected Walking for Mixed Reality Mutual Space Generation.” *IEEE VR*, 2021.
- [C.1] J.-e. Shin, B. Yoon, **Dooyoung Kim**, W. Woo. “A User-Oriented Approach to Space-Adaptive Augmentation: The Effects of Spatial Affordance on Narrative Experience in an Augmented Reality Detective Game.” *ACM CHI*, 2021. ★ **Honorable Mention Award (Top 5%)**

#### *Refereed workshop, adjunct, and demo papers*

- [W.11] **Dooyoung Kim**, W. Woo. “Spatio-Temporal Mixed and Augmented Reality Experience Description for Interactive Playback.” *IEEE ISMAR-Adjunct*, 2025.
- [W.10] **Dooyoung Kim**, K. Kim, C. T. Silva, Q. Sun, D. Turakhia, Z. Wang, K. Wang, K. Perlin, S. Feiner, W. Woo. “An Overview of the 1st International Workshop on Spatial Memory in XR: The Future of Memory Capture and Replay Through XR and AI (XRMemory).” *IEEE VRW*, 2025.
- [W.9] S. Kim, **Dooyoung Kim**, T. Son, W. Woo. “RealityCrafter: User-Guided Editable 3D Scene Generation from a Single Image in Mixed Reality.” *ACM UIST-Adjunct*, 2025. *(co-advised, student first author)*
- [W.8] T. Ha, S. Choi, S. Kim, **Dooyoung Kim**, W. Woo. “Human-Scene Interaction Data Generation with Virtual Environment using User-Centric Scene Graph.” *IEEE VRW*, 2025. *(co-advised, student first author)*
- [W.7] S. Kim, **Dooyoung Kim**, J.-e. Shin, W. Woo. “Object Cluster Registration of Dissimilar Rooms using Geometric Spatial Affordance Graph to Generate Shared Virtual Spaces.” *IEEE VR*, 2024. *(co-advised, student first author)*
- [W.6] J. Kim, **Dooyoung Kim**, B. Kim, H. Kim, J. Lee. “Holobot: Hologram-Based Extended Reality Telepresence Robot.” *ACM/IEEE HRI Companion*, 2023.
- [W.5] **Dooyoung Kim**, H.-i. Kim, W. Woo. “Mutual Space Generation with Relative Translation Gains in Redirected Walking for Asymmetric Remote Collaboration.” *IEEE ISMAR-Adjunct*, 2022.
- [W.4] J. Du, S. Park, J. Yum, Z. Ö. Özdemir, **Dooyoung Kim**, S. Y. Oh, S. H. Yoon. “Art Rich: Place Your AR Artwork.” *IEEE ISMAR-Adjunct*, 2022.
- [W.3] S. Kim, H. Jang, K. T. Oh, S. Y. Oh, **Dooyoung Kim**, W. Woo, J. Lee, J. Ahn, S. H. Yoon. “Bring Store in My Room: AR Store Authoring System for Spatial Experience in Mobile Shopping.” *IEEE ISMAR-Adjunct*, 2022.
- [W.2] S. U. Lee, J. Yun, D. Kim, **Dooyoung Kim**, S. Y. Oh, S. H. Yoon. “CARDS: Comprehensive AR Docent System.” *IEEE ISMAR-Adjunct*, 2022.
- [W.1] H.-I. Kim, T. Kim, E. Song, S. Y. Oh, **Dooyoung Kim**, W. Woo. “Multi-Scale Mixed Reality Collaboration for Digital Twin.” *IEEE ISMAR-Adjunct*, 2021.

#### **PATENTS AND STANDARDISATION**

##### *Issued (granted)*

- KR/US **Virtual Reality Space Adjusting Method with Relative Translation Gain in Redirected Walking and the System.**
- KR/PCT **Method and System for Generating Remote Collaboration Mutual Space.**
- KR **Method of Operating Mixed Reality Telepresence System and Mixed Reality Telepresence System.**
- KR **Method and System for Processing Mixed and Augmented Reality Experiences.**
- KR **Image Processing Device and Recording Medium for Storing Video Processed Therefrom.**

### *Filed / pending*

- KR/US Edge-Centric Space Rescaling Method for Dissimilar Space Registration and the System Thereof.
- KR Spatial Extension Using Human Stereoscopic Perception with a Large Wall Display.
- KR A System for Real-Time Affordance Visualization and Interaction with Meta-Objects based on Dynamic User Context.
- KR Object Cluster Registration of Dissimilar Rooms Using Geometric Spatial Affordance Graph to Generate Shared Virtual Spaces.

### *International standardisation*

- 2025 ISO/IEC JTC 1/SC 24 — Approval of Preliminary Work Item (PWI): *Spatio-Temporal Mixed and Augmented Reality Experience Description (MAR-ED) for Adaptive Playback*.

### **INVITED TALKS**

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- July 2026 Monash University (host: Prof. Tim Dwyer; Department of Human Centred Computing) — Connecting Intelligences beyond Boundaries.
- May 2026 University of Adelaide (host: APrf. Gun Lee; Australian Research Centre for Interactive and Virtual Environments) — Interactive XR Intelligence.
- May 2026 Handong Global University (host: Prof. Minkyu Ahn; School of Artificial Intelligence, Computer and Electrical Engineering) — XRMemory and Physical AI.
- 2024 Korea Computer Congress (KCC), Top-Conference Session.
- 2024 University of Minnesota (host: Prof. Victoria Interrante) — Redirected Walking & Spatial AI.
- 2023 Korea Computer Congress (KCC), Top-Conference Session.
- 2023 Korea Software Congress (KSC), Top-Conference Session.
- 2023 HCI Korea Conference — Metaverse Research and Development Session.
- 2022 ACM SIGGRAPH Asia — OpenXR BoF Session.
- 2022 Asia-Pacific Workshop on Mixed and Augmented Reality (APMAR), Pitch-Your-Work Session.

### **CONFERENCE AND EDITORIAL SERVICE**

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#### *Editorial*

- 2025– **Lead Guest Editor**, Special Issue on “XRMemory: The Future of Memory Capture, Reconstruction, and Replay in XR,” Springer *Virtual Reality Journal* (Q1, IF 5.4).
- 2025– **Guest Editor**, Special Issue on “XR for Knowledge Work and Workplace Enhancement: Human-Centered, Productive, and Ethical XR Workspaces,” Springer *Journal of Ambient Intelligence and Humanized Computing*.

#### *Organising committee — IEEE ISMAR / VR (flagship XR venues)*

- 2026 **Doctoral Consortium Chair**, IEEE VR 2027.
- 2026 **Sponsorship Chair**, IEEE ISMAR 2026.
- 2026 **Standardization Committee**, IEEE VR 2026.
- 2026 **Principal Organiser**, 4th International XRMemory Workshop @ IEEE ISMAR 2026.
- 2026 **Principal Organiser**, 3rd International XRMemory Workshop @ IEEE VR 2026.
- 2025 **Assistant to General Chair & Social Event Chair**, IEEE ISMAR 2025 Organising Committee.
- 2025 **Principal Organiser**, 2nd International XRMemory Workshop @ IEEE ISMAR 2025.
- 2025 **Organiser**, XRWorks’25 Workshop @ IEEE ISMAR 2025; IPC, XRStand Workshop.
- 2025 **Principal Organiser**, 1st International XRMemory Workshop @ IEEE VR 2025.



### *Organising committee — other*

- 2025    Organiser & Program Chair, 3rd International Symposium on Meta-AI for X (IS-MAX).
- 2025    Organiser, MetaMuseum Workshop @ HCI Korea Conference.
- 2024    Poster Chair (Computer Graphics & Interaction); Metaverse+AI+XR Workshop Organiser — Korea Computer Congress (KCC).
- 2024    Poster Chair (CG/HCI/AI/Metaverse/AR/VR) — Korea Software Congress (KSC).
- 2024    Organiser & Web Chair, 2nd International Symposium on Meta-AI for X (ISMAX).
- 2023    Organiser & Student Chair, 1st International Symposium on Meta-AI for X (IS-MAX).

### *International Program Committee (IPC) & Reviewer*

- IEEE ISMAR    2026 (IPC), 2025 (IPC), 2024, 2023, 2022, 2021.
- IEEE VR       2027 (IPC), 2026 (IPC), 2025 (IPC), 2024, 2023, 2022.
- ACM VRST     2026 (IPC), 2025 (IPC), 2024 (IPC).
- ACM CHI       2026, 2025, 2024, 2023.
- IEEE/ACM HRI   2023.

## **TEACHING**

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### *Lecturer — La Trobe*

- 2026    **CSE2002** — Agentic AI Software Engineering; developing a new course.

### *Graduate course co-instructor — KAIST*

- 2025    **ARP700** — AR project mentoring; teaching assistant on Spatial AI & XRMemory.
- 2024    **ARP700** — AR project mentoring; teaching assistant on XR & Spatial Computing.
- 2024    **MV600** — AR/Metaverse project mentoring; teaching assistant on MR telepresence.
- 2023    **CTP445** — AR fundamentals; discussion sessions; teaching assistant on SLAM with OpenCV.
- 2022    **GCT555** — Unity & AR/VR for graduate students; teaching assistant on HoloLens 2, MRTK with Unity.
- 2022    **CTP445** — Unity & AR/VR for graduate students; teaching assistant on smartphone, ARCore with Unity).

## **COMMUNITY AND OUTREACH**

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- Jan 2023 – May 2026    **Global Shaper**, World Economic Forum. Founding Curator of Daejeon Hub; Vice Curator (2023–2024).
- Mar 2022 – Dec 2022    Instructor, *Multi-User Game Development with Unity*, KAIST Center for Gifted Education.
- Jul 2021 – Aug 2021    Python Coach, NAVER.